EXPERIMENT 10:

PHOTODIODE RESPONSE AND SENSITIVITY ANALYSIS USING OPTISYSTEM

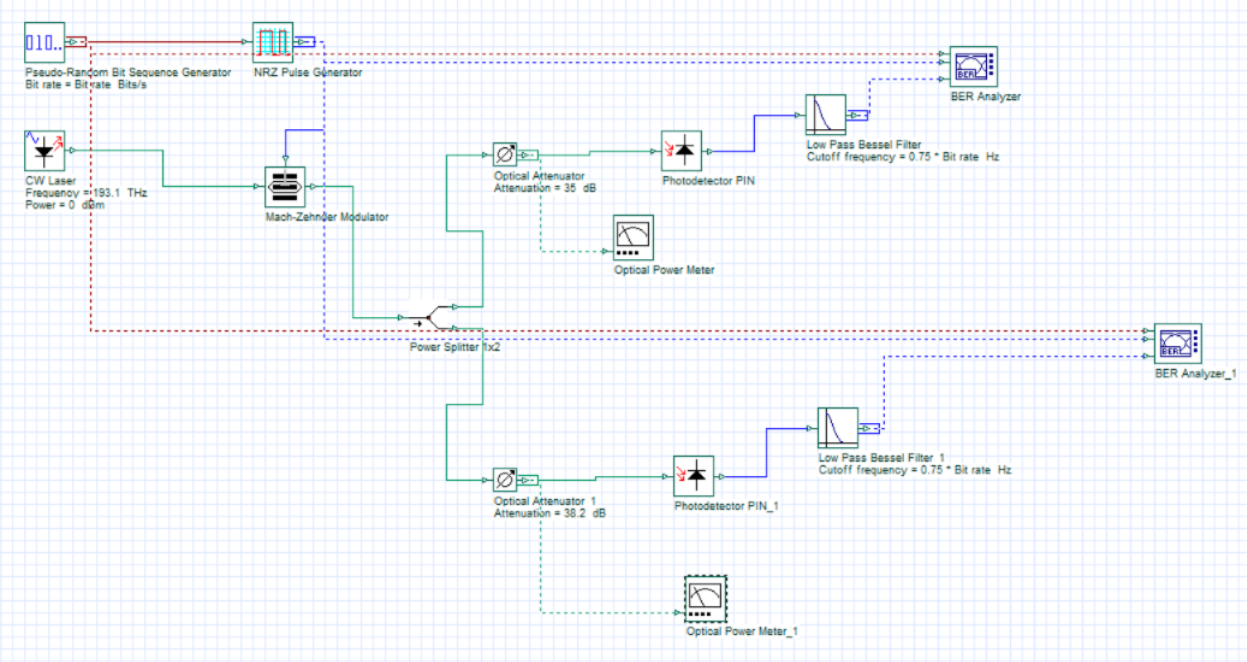
Objective :

To design and simulate an optical communication receiver using OptiSystem, analyze the performance of PIN photodiodes under different attenuation levels, and compare the received optical power and receiver sensitivity using optical power meters and BER analyzers.

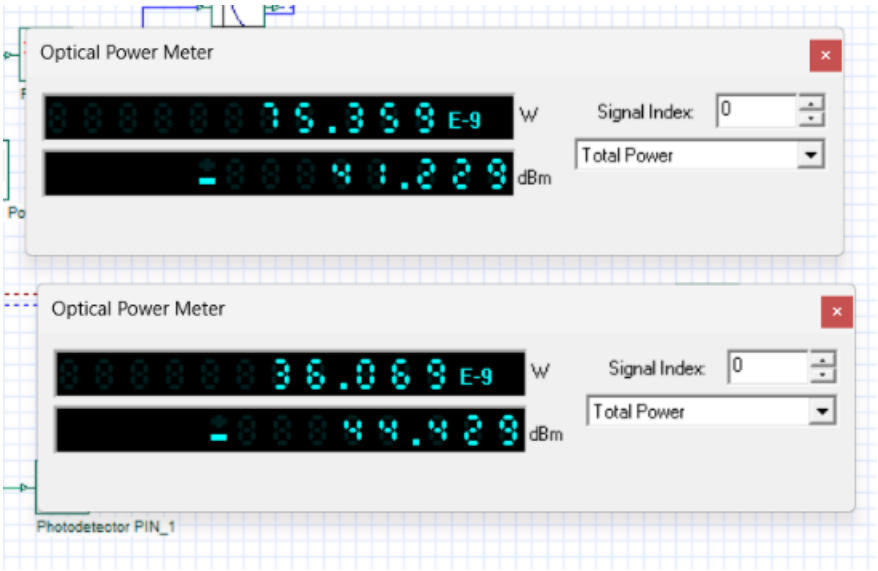
SOFTWARE REQUIRED:

OptiSystem.

circuit DIAGRAM:



OUTPUTS:



REFLECTION:

- I learned how a PIN photodiode converts optical signals into electrical signals.
- I understood how attenuation reduces the received optical power.
- I learned to use optical power meters to measure received power.
- I understood the importance of BER analysis in evaluating receiver performance

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RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improve-ment – 1	Weight-age
Design the OptiSystem PIN/APD receiver circuit with correct components and parameters.	Circuit Setup and Parameter Definition	Builds the complete circuit (laser, modulator, Fork 1×2, attenuators, PIN/APD, filters, power meters, BER analyzers) with all parameters correct (193.1 THz, 0 dBm, 35/38.2 dB, 0.75 × bit rate), independently.	Builds the circuit with minor connection or parameter errors and limited assistance.	Builds only part of the circuit or needs considerable assistance; has unit/parameter errors (e.g., dB vs dBm).	Unable to build the circuit or set the parameters.	25
Run the simulation and record received power, Q factor and BER for both receivers.	Simulation Execution and Data Acquisition	Runs the simulation and records received power (W and dBm), eye diagram, Q factor, BER and the attenuation sweep correctly for PIN and APD.	Records all quantities with minor errors or omissions.	Records some quantities with noticeable errors or inconsistencies.	Unable to run the simulation or record the values.	25
Obtain eye diagrams and BER/SNR plots; determine responsivity, sensitivity and bandwidth.	Graph Plotting and Sensitivity Determination	Plots accurate eye diagrams and BER (semi-log) / SNR curves with labelled axes and a BER = 10^{-9} line; sensitivity within ±1 dB, responsivity and bandwidth within 10% of the reference.	Plots the curves with minor errors; sensitivity, responsivity and bandwidth identified with small deviation.	Plots partial or poorly scaled curves; sensitivity, responsivity or bandwidth approximate.	Unable to plot the curves or determine the parameters.	25
Interpret photocurrent–power relation, SNR/BER variation and PIN vs APD performance.	Analysis and Engineering Interpretation	Correctly explains linear photocurrent, noise-limited regions, SNR/BER trend and the APD gain advantage, and links results to detector selection.	Explains the trends with minor errors and reasonable interpretation.	Performs basic interpretation with noticeable errors or limited justification.	Unable to interpret the results.	25

CONCLUSION :

1. The optical receiver system was designed and simulated successfully using OptiSystem.
2. Two receiver branches with different attenuation values were analyzed.
3. The optical power meter readings showed that the received power decreased as attenuation increased.
4. The BER analyzers were used to evaluate the performance of both receiver branches.
5. The experiment helped demonstrate the effect of optical attenuation on photodiode receiver performance and sensitivity.